

1 Bounded Error of Implicit Tucker Self-Expression

This section proves that replacing the explicit multi-view self-expression coefficient tensor with the implicit Tucker parameterization yields a bounded reconstruction error. The final error consists of four terms: ideal self-expression residual, Tucker truncation error, optimization residual, and S-PMA approximation error.

Therefore, as long as the reference self-expression tensor is approximately low Tucker-rank and the S-PMA approximation error is controlled, the implicit Tucker self-expression used by DMTSE does not cause the reconstruction error to become unbounded.

Notation. Let the multi-view traffic features be

$$\{X^{(k)}\}_{k=1}^K, \quad X^{(k)} \in \mathbb{R}^{d_k \times N},$$

where N is the number of flows and K is the number of views. Let

$$\mathcal{Z} \in \mathbb{R}^{N \times N \times K}$$

denote the reference full self-expression coefficient tensor. Its k -th view slice is denoted by

$$Z^{(k)} \in \mathbb{R}^{N \times N}.$$

The word “full” means that the tensor covers all flow pairs and all views; it does not mean that its entries are independent. In traffic data, many flows are generated by a small number of latent behavior patterns, so the full coefficient tensor can still have an approximately low Tucker-rank structure.

DMTSE uses the implicit Tucker parameterization

$$\widehat{\mathcal{Z}} = \mathcal{G} \times_1 A \times_2 B \times_3 \Gamma,$$

where $A \in \mathbb{R}^{N \times P}$, $B \in \mathbb{R}^{N \times Q}$, $\Gamma \in \mathbb{R}^{K \times R}$, and $\mathcal{G} \in \mathbb{R}^{P \times Q \times R}$.

Let γ_k be the k -th row of Γ , and define

$$M_k = \mathcal{G} \times_3 \gamma_k = \sum_{r=1}^R \gamma_{kr} \mathcal{G}_{::r} \in \mathbb{R}^{P \times Q}.$$

Then the implicit coefficient matrix of view k is

$$\widehat{Z}^{(k)} = A M_k B^\top.$$

Let $\phi : \mathbb{R} \rightarrow \mathbb{R}$ be the elementwise output activation in the reconstruction layer. We assume that ϕ is L_ϕ -Lipschitz:

$$|\phi(a) - \phi(b)| \leq L_\phi |a - b|.$$

For the sigmoid activation, one can take $L_\phi = 1/4$; for the identity map, one can take $L_\phi = 1$.

Spectral energy ratio. Let $\mathcal{Z}_{(n)}$ be the mode- n unfolding of $\mathcal{Z}$, and let its singular values be sorted in non-increasing order:

$$\sigma_1(\mathcal{Z}_{(n)}) \geq \sigma_2(\mathcal{Z}_{(n)}) \geq \cdots.$$

For a target multilinear rank $(r_1, r_2, r_3) = (P, Q, R)$, define the cumulative spectral-energy ratio of the n -th unfolding as

$$g_n(r_n) = \frac{\sum_{i=1}^{r_n} \sigma_i^2(\mathcal{Z}_{(n)})}{\sum_i \sigma_i^2(\mathcal{Z}_{(n)})}.$$

Since tensor unfolding preserves the Frobenius norm,

$$\sum_i \sigma_i^2(\mathcal{Z}_{(n)}) = \|\mathcal{Z}\|_F^2, \quad n = 1, 2, 3.$$

Define the Tucker truncation error

$$\varepsilon_{\text{Tuc}} = \|\mathcal{Z}\|_F \sqrt{\sum_{n=1}^3 (1 - g_n(r_n))}, \quad (r_1, r_2, r_3) = (P, Q, R).$$

Definition 1 (Optimization gap to the HOSVD witness). *Let $\widehat{\mathcal{Z}}^{\text{HO}}$ be the rank- (P, Q, R) truncated-HOSVD approximation of $\mathcal{Z}$, and let $\widehat{\mathcal{Z}}$ be the Tucker tensor learned by DMTSE. Define the optimization gap*

$$\varepsilon_{\text{opt}} := \|\widehat{\mathcal{Z}} - \widehat{\mathcal{Z}}^{\text{HO}}\|_F,$$

i.e. the Frobenius distance between the learned factors and the truncated-HOSVD witness of the reference tensor. This term is small whenever training converges to a factorization close to the best rank- (P, Q, R) approximation; it isolates the (non-convex) optimization error from the representational error analyzed below.

Theorem 1 (Bounded reconstruction error of implicit Tucker self-expression). *Under Definition 1, define the ideal self-expression residual of view k as*

$$\xi_k = \left\| X^{(k)} - \phi\left(X^{(k)} Z^{(k)}\right) \right\|_F.$$

Let

$$U^{(k)} = X^{(k)} A$$

be the exact feature-side Tucker embedding, and let

$$\widetilde{U}^{(k)}$$

be the approximate embedding computed by S-PMA and the view-specific refiner. Define

$$\eta_k = \left\| \widetilde{U}^{(k)} - U^{(k)} \right\|_F.$$

The final DMTSE reconstruction of view k is

$$\widetilde{X}^{(k)} = \phi\left(\widetilde{U}^{(k)} M_k B^\top\right).$$

Define the total reconstruction error

$$\mathcal{E}_{\text{rec}} = \left(\sum_{k=1}^K \left\| X^{(k)} - \widetilde{X}^{(k)} \right\|_F^2 \right)^{1/2},$$

the total ideal self-expression residual

$$\varepsilon_{\text{se}} = \left(\sum_{k=1}^K \xi_k^2 \right)^{1/2},$$

the data spectral-norm bound

$$\rho_X = \max_{1 \leq k \leq K} \|X^{(k)}\|_2,$$

the Tucker view-map bound

$$\rho_M = \max_{1 \leq k \leq K} \|M_k\|_2,$$

and the total S-PMA approximation error

$$\varepsilon_{\text{PMA}} = \left(\sum_{k=1}^K \eta_k^2 \right)^{1/2}.$$

Then

$$\mathcal{E}_{\text{rec}} \leq \varepsilon_{\text{se}} + L_\phi \rho_X (\varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}) + L_\phi \rho_M \|B\|_2 \varepsilon_{\text{PMA}}.$$

Here

$$\varepsilon_{\text{Tuc}} = \|\mathcal{Z}\|_F \sqrt{\sum_{n=1}^3 (1 - g_n(r_n))}.$$

Therefore, when the spectral energy of the three unfoldings of $\mathcal{Z}$ concentrates in the target ranks, the optimization gap ε_{opt} is small, and the S-PMA embedding error is controlled, the implicit Tucker reconstruction error is bounded.

Proof. We restate Theorem 1 in Sec VII.A with full definitions and prove it here.

Step 1: Tucker truncation error is controlled by tail spectral energy. By the standard truncated-HOSVD approximation bound, there exists a rank- (P, Q, R) Tucker tensor $\hat{\mathcal{Z}}^{\text{HO}}$ such that

$$\left\| \mathcal{Z} - \hat{\mathcal{Z}}^{\text{HO}} \right\|_F^2 \leq \sum_{n=1}^3 \sum_{i > r_n} \sigma_i^2(\mathcal{Z}_{(n)}).$$

By the definition of $g_n(r_n)$,

$$\sum_{i > r_n} \sigma_i^2(\mathcal{Z}_{(n)}) = (1 - g_n(r_n)) \|\mathcal{Z}\|_F^2.$$

Hence

$$\left\| \mathcal{Z} - \hat{\mathcal{Z}}^{\text{HO}} \right\|_F^2 \leq \|\mathcal{Z}\|_F^2 \sum_{n=1}^3 (1 - g_n(r_n)).$$

Taking square roots gives

$$\left\| \mathcal{Z} - \hat{\mathcal{Z}}^{\text{HO}} \right\|_F \leq \varepsilon_{\text{Tuc}}.$$

Step 2: The optimization gap gives an explicit tensor-distance bound. By Definition 1,

$$\|\widehat{\mathcal{Z}} - \widehat{\mathcal{Z}}^{\text{HO}}\|_F = \varepsilon_{\text{opt}}.$$

By the triangle inequality and Step 1,

$$\|\mathcal{Z} - \widehat{\mathcal{Z}}\|_F \leq \|\mathcal{Z} - \widehat{\mathcal{Z}}^{\text{HO}}\|_F + \|\widehat{\mathcal{Z}}^{\text{HO}} - \widehat{\mathcal{Z}}\|_F \leq \varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}.$$

Let $\Delta_{\mathcal{Z}} = \mathcal{Z} - \widehat{\mathcal{Z}}$. Then

$$\|\Delta_{\mathcal{Z}}\|_F \leq \varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}.$$

Step 3: Coefficient-tensor error propagates to exact Tucker reconstruction. For each view k , define

$$\Delta_k = Z^{(k)} - \widehat{Z}^{(k)}.$$

Since the Frobenius norm of a tensor equals the square root of the sum of the squared Frobenius norms of its slices,

$$\sum_{k=1}^K \|\Delta_k\|_F^2 = \|\mathcal{Z} - \widehat{\mathcal{Z}}\|_F^2 \leq (\varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}})^2.$$

Consider the reconstruction obtained from the learned Tucker coefficient matrix but without the S-PMA approximation:

$$\widehat{X}_{\text{exact}}^{(k)} = \phi\left(X^{(k)} \widehat{Z}^{(k)}\right).$$

By the triangle inequality,

$$\begin{aligned} \|X^{(k)} - \widehat{X}_{\text{exact}}^{(k)}\|_F &= \|X^{(k)} - \phi\left(X^{(k)} \widehat{Z}^{(k)}\right)\|_F \\ &\leq \|X^{(k)} - \phi\left(X^{(k)} Z^{(k)}\right)\|_F \\ &\quad + \|\phi\left(X^{(k)} Z^{(k)}\right) - \phi\left(X^{(k)} \widehat{Z}^{(k)}\right)\|_F. \end{aligned}$$

The first term is ξ_k . Since ϕ is L_ϕ -Lipschitz,

$$\begin{aligned} &\|\phi\left(X^{(k)} Z^{(k)}\right) - \phi\left(X^{(k)} \widehat{Z}^{(k)}\right)\|_F \\ &\leq L_\phi \|X^{(k)} (Z^{(k)} - \widehat{Z}^{(k)})\|_F \\ &\leq L_\phi \|X^{(k)}\|_2 \|Z^{(k)} - \widehat{Z}^{(k)}\|_F \\ &= L_\phi \|X^{(k)}\|_2 \|\Delta_k\|_F. \end{aligned}$$

Therefore,

$$\|X^{(k)} - \widehat{X}_{\text{exact}}^{(k)}\|_F \leq \xi_k + L_\phi \|X^{(k)}\|_2 \|\Delta_k\|_F.$$

Step 4: S-PMA embedding error propagates to the final reconstruction. By the implicit Tucker structure,

$$\widehat{Z}^{(k)} = A M_k B^\top.$$

Therefore,

$$X^{(k)} \widehat{Z}^{(k)} = X^{(k)} A M_k B^\top = U^{(k)} M_k B^\top.$$

The actual model uses the S-PMA approximation $\tilde{U}^{(k)}$, so

$$\tilde{X}^{(k)} = \phi\left(\tilde{U}^{(k)} M_k B^\top\right).$$

Thus,

$$\begin{aligned} & \left\| \hat{X}_{\text{exact}}^{(k)} - \tilde{X}^{(k)} \right\|_F \\ &= \left\| \phi\left(U^{(k)} M_k B^\top\right) - \phi\left(\tilde{U}^{(k)} M_k B^\top\right) \right\|_F \\ &\leq L_\phi \left\| \left(U^{(k)} - \tilde{U}^{(k)}\right) M_k B^\top \right\|_F \\ &\leq L_\phi \left\| U^{(k)} - \tilde{U}^{(k)} \right\|_F \|M_k\|_2 \|B\|_2 \\ &= L_\phi \eta_k \|M_k\|_2 \|B\|_2. \end{aligned}$$

Combining Step 3 and Step 4 gives, for each view k ,

$$\left\| X^{(k)} - \tilde{X}^{(k)} \right\|_F \leq \xi_k + L_\phi \|X^{(k)}\|_2 \|\Delta_k\|_F + L_\phi \eta_k \|M_k\|_2 \|B\|_2.$$

Taking the squared sum over all views and applying Minkowski's inequality,

$$\begin{aligned} \mathcal{E}_{\text{rec}} &= \left(\sum_{k=1}^K \left\| X^{(k)} - \tilde{X}^{(k)} \right\|_F^2 \right)^{1/2} \\ &\leq \left(\sum_{k=1}^K \xi_k^2 \right)^{1/2} \\ &\quad + L_\phi \left(\sum_{k=1}^K \|X^{(k)}\|_2^2 \|\Delta_k\|_F^2 \right)^{1/2} \\ &\quad + L_\phi \|B\|_2 \left(\sum_{k=1}^K \eta_k^2 \|M_k\|_2^2 \right)^{1/2}. \end{aligned}$$

By definition,

$$\left(\sum_{k=1}^K \xi_k^2 \right)^{1/2} = \varepsilon_{\text{se}}.$$

Since $\rho_X = \max_k \|X^{(k)}\|_2$,

$$\left(\sum_{k=1}^K \|X^{(k)}\|_2^2 \|\Delta_k\|_F^2 \right)^{1/2} \leq \rho_X \left(\sum_{k=1}^K \|\Delta_k\|_F^2 \right)^{1/2}.$$

Using Step 2,

$$\left(\sum_{k=1}^K \|\Delta_k\|_F^2 \right)^{1/2} = \|\mathcal{Z} - \hat{\mathcal{Z}}\|_F \leq \varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}.$$

Furthermore, since $\rho_M = \max_k \|M_k\|_2$,

$$\left(\sum_{k=1}^K \eta_k^2 \|M_k\|_2^2 \right)^{1/2} \leq \rho_M \left(\sum_{k=1}^K \eta_k^2 \right)^{1/2} = \rho_M \varepsilon_{\text{PMA}}.$$

Therefore,

$$\mathcal{E}_{\text{rec}} \leq \varepsilon_{\text{se}} + L_{\phi}\rho_X (\varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}) + L_{\phi}\rho_M \|B\|_2 \varepsilon_{\text{PMA}}.$$

Substituting the expression of ε_{Tuc} completes the proof. $\square$

Corollary 1 (Explicit bound induced by per-position S-PMA error). *Suppose S-PMA satisfies, for every view k and every feature position m ,*

$$\left\| \tilde{u}_m^{(k)} - u_m^{(k)} \right\|_2 \leq C_{\text{spma}}^{(k)}.$$

Then

$$\eta_k = \left\| \tilde{U}^{(k)} - U^{(k)} \right\|_F \leq \sqrt{d_k} C_{\text{spma}}^{(k)}.$$

Therefore,

$$\varepsilon_{\text{PMA}} \leq \left(\sum_{k=1}^K d_k \left(C_{\text{spma}}^{(k)} \right)^2 \right)^{1/2}.$$

If all views share a common bound C_{spma} , then

$$\varepsilon_{\text{PMA}} \leq C_{\text{spma}} \sqrt{\sum_{k=1}^K d_k}.$$

Substituting it into Theorem 1 gives

$$\mathcal{E}_{\text{rec}} \leq \varepsilon_{\text{se}} + L_{\phi}\rho_X (\varepsilon_{\text{Tuc}} + \varepsilon_{\text{opt}}) + L_{\phi}\rho_M \|B\|_2 C_{\text{spma}} \sqrt{\sum_{k=1}^K d_k}.$$

Proof. By the definition of the Frobenius norm,

$$\eta_k^2 = \left\| \tilde{U}^{(k)} - U^{(k)} \right\|_F^2 = \sum_{m=1}^{d_k} \left\| \tilde{u}_m^{(k)} - u_m^{(k)} \right\|_2^2.$$

If

$$\left\| \tilde{u}_m^{(k)} - u_m^{(k)} \right\|_2 \leq C_{\text{spma}}^{(k)},$$

then

$$\eta_k^2 \leq d_k \left(C_{\text{spma}}^{(k)} \right)^2,$$

which gives

$$\eta_k \leq \sqrt{d_k} C_{\text{spma}}^{(k)}.$$

Summing over all views yields

$$\varepsilon_{\text{PMA}} = \left(\sum_{k=1}^K \eta_k^2 \right)^{1/2} \leq \left(\sum_{k=1}^K d_k \left(C_{\text{spma}}^{(k)} \right)^2 \right)^{1/2}.$$

The shared-bound case follows immediately by setting $C_{\text{spma}}^{(k)} = C_{\text{spma}}$. Substitution into Theorem 1 completes the proof. $\square$

Remark 1 (Interpretation). *Theorem 1 shows that the final reconstruction error of DMTSE is controlled by four terms:*

$$\varepsilon_{\text{se}}, \quad \varepsilon_{\text{Tuc}}, \quad \varepsilon_{\text{opt}}, \quad \varepsilon_{\text{PMA}}.$$

The term ε_{se} measures how well the original data can be explained by an ideal full self-expression model. The term ε_{Tuc} is determined by the tail singular-value energy of the three unfoldings of the full coefficient tensor, and is directly supported by the mode-wise spectral-energy measurements. The term ε_{opt} makes the optimization gap to the truncated-HOSVD witness explicit. The term ε_{PMA} measures the approximation error introduced when S-PMA replaces the full feature-side aggregation $U^{(k)} = X^{(k)}A$. Therefore, if the traffic self-expression tensor has a clear low Tucker-rank structure and S-PMA gives a controlled approximation, the implicit Tucker generator can avoid materializing the $N \times N \times K$ tensor while maintaining a bounded reconstruction error.